COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 2 : Session – 5 : SLO – 2 | Memory Operations and Byte Addressing
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

Section A – Match the Following (Answers)

• 1. Load Operation → E. Memory read from a specified address
• 2. Store Operation → B. Memory write from processor to specified location
• 3. Big-Endian → D. Most significant byte stored at lowest address
• 4. Little-Endian → A. Least significant byte stored at lowest address
• 5. 32-bit Data Alignment → C. Word starts at byte address 0, 4, 8…

Section B – Fill in the Blanks

• The LOAD operation reads the contents of a specified memory location.
• The STORE operation stores processor data into a specific memory location.
• In BIG-ENDIAN representation, the most significant byte is stored at the lowest memory address.
• In LITTLE-ENDIAN representation, the least significant byte is stored at the lowest memory address.
• In 64-bit data alignment, words begin at byte addresses such as 0, 8, 16, and so on.

Section C – Activity Task

• Task 1 – Word 0x46789654 stored at address 8000h:
• (a) Big-endian: 8000 → 46, 8001 → 78, 8002 → 96, 8003 → 54.
• (b) Little-endian: 8000 → 54, 8001 → 96, 8002 → 78, 8003 → 46.
• Task 2a – Why endianness matters: machines that use different byte orders interpret the same binary sequence differently; when binary data is transferred between systems the byte order must be agreed or converted, otherwise multi-byte values are read incorrectly.
• Task 2b – Aligned vs unaligned access: an aligned word starts at an address that is a multiple of its size (e.g. 0, 4, 8 for 4-byte words) and is fetched in one memory cycle; an unaligned word straddles two word boundaries, needing multiple accesses and extra merging logic, so it is slower and on some architectures not permitted.

Section D – True or False

• In Big-endian format, the most significant byte is stored last. — FALSE (it is stored at the lowest address)
• Aligned 16-bit words begin at addresses like 0, 2, 4… — TRUE
• Little-endian format is commonly used in Intel processors. — TRUE
• A load operation writes data to memory. — FALSE (load reads memory; store writes)
• Word alignment helps in faster memory access and avoids errors in some architectures. — TRUE